

Kaustav Purkayastha

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🌐 [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SKILLS

Languages & Data: Python, SQL, PySpark, PL/SQL, Shell Scripting, ETL/ELT, CDC, SCD Type-2, Data Warehousing, Data Quality

Cloud & Platforms: AWS (Redshift, S3, Glue, EC2, Aurora), Snowflake, Databricks, dbt, Informatica IDMC, PostgreSQL, Oracle

Tools & DevOps: Stonebranch, Apache Airflow, Git/GitHub, Docker, Tableau Prep, Amazon Bedrock, Anthropic Claude

WORK EXPERIENCE

Deloitte Consulting, Hyderabad Oct 2023 – Present
Data Engineer – AI & Data Practice

- Architected a **centralized data platform** for a Fortune 500 insurance client's Retail Annuities LOB, engineering **70+ Spark-based ETL jobs** in Informatica IDMC and loading **60+ Redshift tables** with 25+ years of historical data using **ELT, CDC, SCD Type-2**, and snapshot ingestion patterns.
- Migrated **4+ legacy Tableau Prep workflows** into scalable **Redshift ELT pipelines**, consolidating multiple RDBMS sources and external datasets, and re-engineered complex Python-based annuity calculations into **optimized SQL**, achieving **~98% data accuracy** against legacy extracts.
- Reverse-engineered **500+ legacy source tables** and authored **100+ Source-to-Target Mapping (STTM)** documents, resolving schema conflicts and mapping ambiguities to unify Annuity and Life data models into a **single integrated warehouse**.
- Designed **50+ parameterized Redshift views** and reusable configuration parameter sets, reducing pipeline configuration overhead across **30+ jobs** and enabling **self-service analytics** for downstream business and reporting teams.
- Integrated **Anthropic Claude with Amazon Bedrock** into the development workflow, accelerating pipeline development by **~50%**, reducing compute costs, and establishing patterns for **AI-assisted code generation** within the data engineering lifecycle.
- Orchestrated end-to-end data ingestion and transformation across **AWS Glue, Redshift, S3, EC2, Aurora**, and **Stonebranch**, while owning patching cycles, **metadata lineage documentation**, QA handoffs, and complex **business rule interpretation**.

PERSONAL PROJECTS

HealthPipe AI [🔗 View Source Code](#) Mar 2026 – May 2026
Multi-agent pipeline that ingests **healthcare data from public APIs**, processes it through **four AI agents**, and loads clean output into **DuckDB**. Includes **quality scorecards**, auto-generated **data dictionaries**, and **PySpark** support for large-scale datasets.
Python, DuckDB, PySpark, Ollama, Pytest, GitHub Actions

TestNucleus [🔗 View Source Code](#) Jun 2025 – Aug 2025
Config-driven **data quality framework** that separates test definitions (**YAML**) from execution logic. Supports **20+ validation checks** across completeness, uniqueness, and referential integrity — built for **CI/CD and pipeline integration**.
Python, SQLAlchemy, Pydantic, Pytest

EDUCATION

Lovely Professional University (LPU) 2024 – 2026
Master of Computer Applications (MCA), Data Science & Finance CGPA: 7.48

Assam University, Silchar 2020 – 2023
Bachelor of Science (Hons.), Computer Science CGPA: 8.86

CERTIFICATIONS & RECOGNITION

- **Certifications:** [Snowflake SnowPro Core \(2026\)](#), [Databricks Certified Data Engineer Associate \(2025\)](#), [Claude with Amazon Bedrock \(2026\)](#), [Industry Proficiency: Financial Services \(2025\)](#), [IDMC Associate \(2024\)](#)
- **Recognition:** Outstanding Performance Award (Feb 2026, May 2025) | Applause Award (Jul 2024) | FY24 Rising Stars (May 2024) | Institutional Merit Scholarship (Dec 2024)